

Given the following distribution:

Calculate: median, mode, mean

Class Interval	[0-10]	[10-20]	[20-30]	[30-40]	[40-50]	[50-60]
Frequency (m_i)	3	5	9	12	7	4
x_i	5	15	25	35	45	55
I.C.F	3	8	17	29	36	40

Solution:

Calculate x_i and I.C.F. , $N = 40$

$$\text{Mean} = \frac{\sum x_i m_i}{N} = \frac{1270}{40} = 31.75$$

$$\text{Median} \xrightarrow{\frac{N}{2}=20} \text{class } [30-40]$$

$$\text{Median} = L + \frac{\frac{N}{2} - \text{I.C.F}_{i-1}}{m_i} \times h$$

$$\text{Median} = 30 + \frac{20 - 17}{12} \times 10 = 32.5$$

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h$$

$$\text{Mode} = 30 + \frac{3}{3 + 5} \times 10 = 33.75$$